



NITI Aayog

Strategy for

New India@75

STRATEGY FOR NEW INDIA @ 75

MODULE - 21

21

WATER RESOURCES

OBJECTIVES

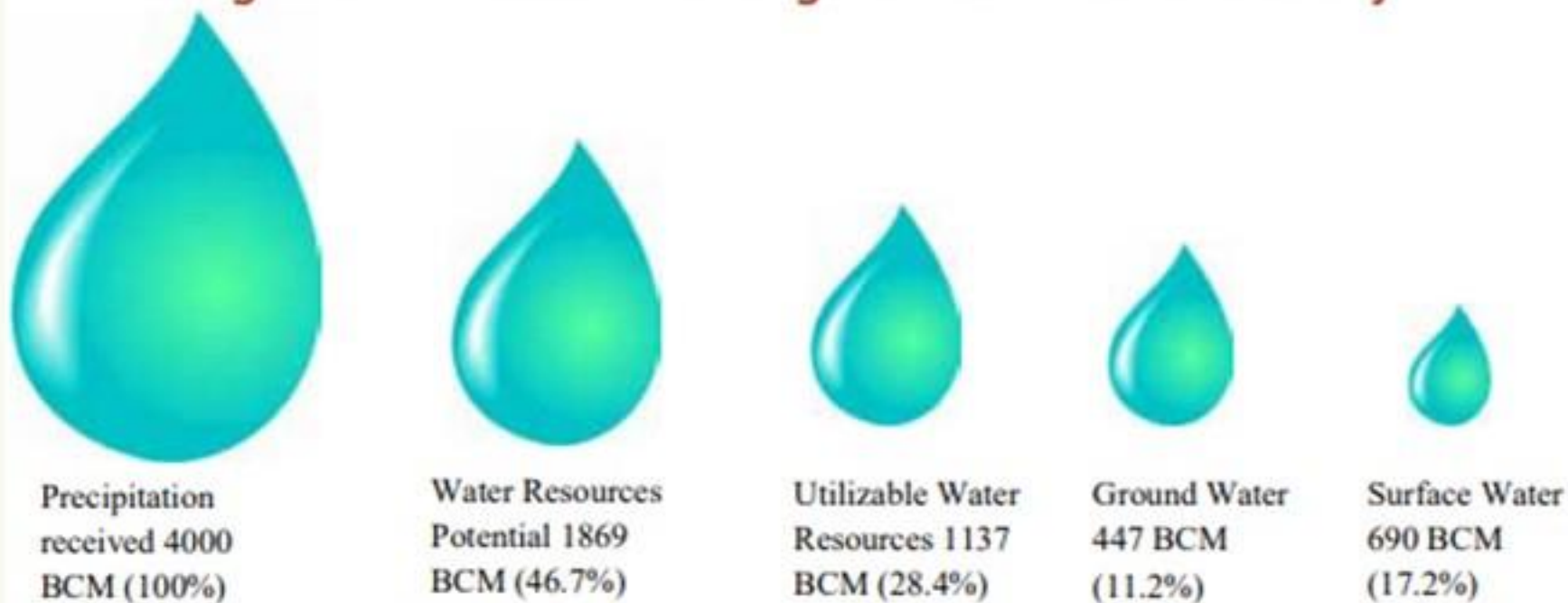


- Provide adequate and safe drinking water (piped) and water for sanitation for citizens and livestock. The minimum water availability should be 40 litres per capita per day in rural areas and 135 litres per capita per day in urban areas.
- Provide irrigation to all farms (Har Khet Ko Pani) with improved on-farm water-use efficiency (more crop per drop).
- Provide water to industries, encourage industries to utilize recycled/treated water and also ensure zero discharge of untreated effluents from industrial units.
- Ensure Aviral and Nirmal Dhara in the Ganga and other rivers.

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WATER RESOURCES

Figure 21.1: Status on average annual water availability



2 ZERO HUNGER



6 CLEAN WATER AND SANITATION



9 INDUSTRY, INNOVATION AND INFRASTRUCTURE



12 RESPONSIBLE CONSUMPTION AND PRODUCTION



14 LIFE BELOW WATER



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WATER RESOURCES

OBJECTIVES



- Create additional water storage capacity to ensure full utilization of the utilizable surface water resources potential of 690 billion cubic metres.
- Ensure long-term sustainability of finite ground water resources.
- Proper operation and maintenance of water infrastructure with active participation of farmers/consumers.
- Promote R&D to facilitate adoption of the latest technologies in the water sector.

PRESENT SCENARIO



Central Water Commission
(Serving the nation since 1945)



Government of India

Ministry of Jal Shakti

Department of Water Resources,
River Development & Ganga Rejuvenation

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- As per Central Water Commission, per capita annual water availability decreased from 1,816 cubic metres in 2001 to 1,544 cubic metres in 2011.
- If the amount of renewable water in a country is below 1,700 cu m per person per year, the country is said to be **experiencing water stress**; below 1,000 cu m, it is said to be **experiencing water scarcity**.



नीति आयोग

National Institution for Transforming India



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WATER RESOURCES

PRESENT SCENARIO

WHY THE WATER RESOURCES ARE FACING PRESSURE?



- Population explosion
- Urbanization
- Rising demand for water from the agriculture
- Energy
- Industry sector
- Pollution
- Inefficient-use
- Poor management
- Poor institutional mechanisms.

Several regions experience water scarcity due to the uneven distribution of water resources over space and time.



PRESENT SCENARIO

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WATER RESOURCES

**PRADHAN MANTRI KRISHI
SINCHAYEE YOJANA**

It is an umbrella scheme for irrigation and prioritized 99 major and medium irrigation projects for completion by December 2019.

**NAMAMI GANGE**

- It is to clean and rejuvenate the River Ganga to maintain “Aviral” and “Nirmal Dhara”.
- This is to ensure its ecological and geological integrity.



Data for 2015-17 indicates improvement in water quality in terms of dissolved oxygen and coliform bacteria. But, a lot more needs to be done to achieve Sewage Treatment Plan targets, and controlling household and industrial waste.

CONSTRAINTS

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WATER RESOURCES

6 The Easement Act, 1882, grants groundwater **ownership rights to the landowner** and it is one of the reasons for water over-use and depletion of groundwater levels.

5 There are also **absence of field channels** (Command Area Development works).

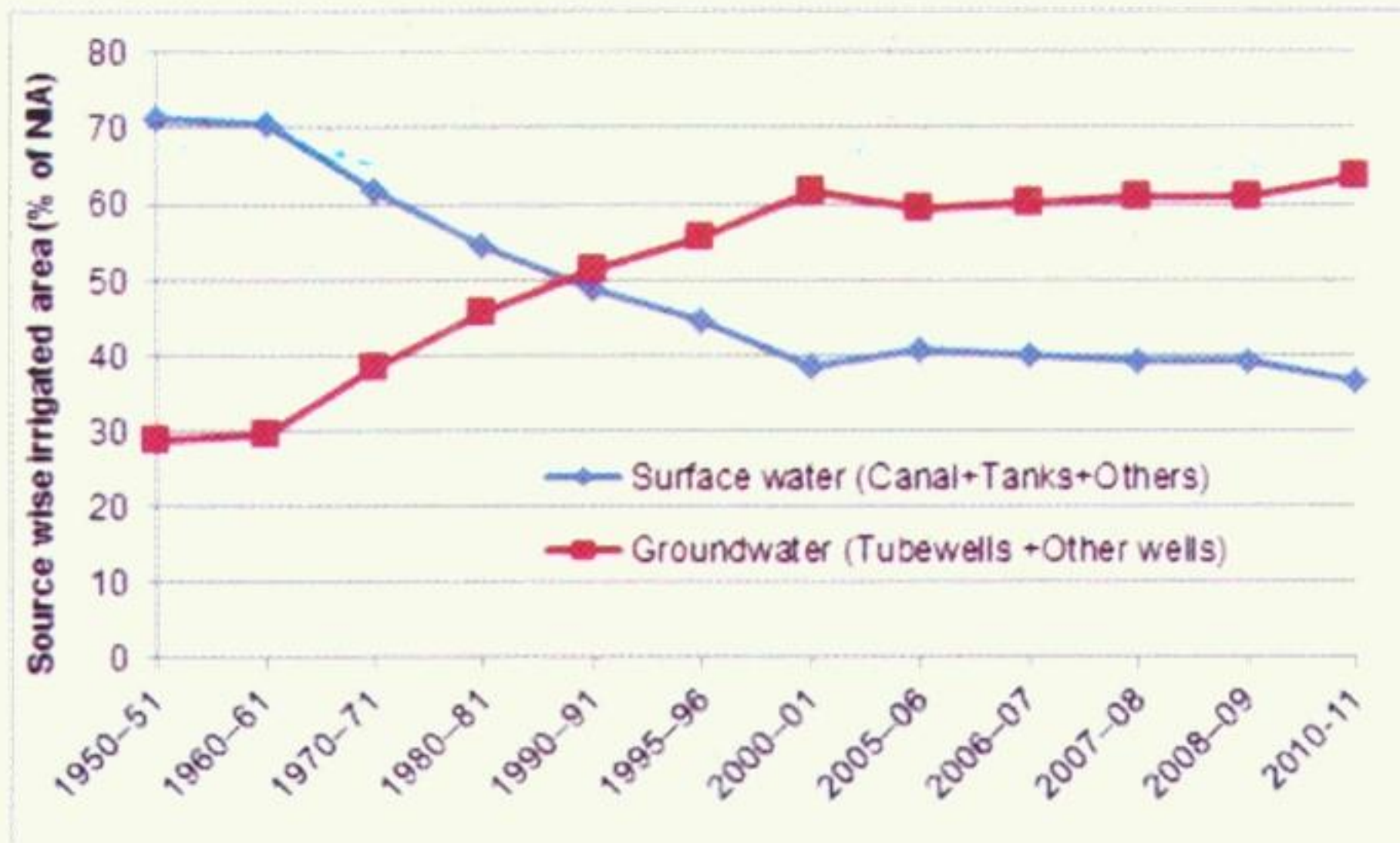
4 There is absence of **participatory irrigation management**, non-alignment of cropping patterns to the **agro-climatic zones**.



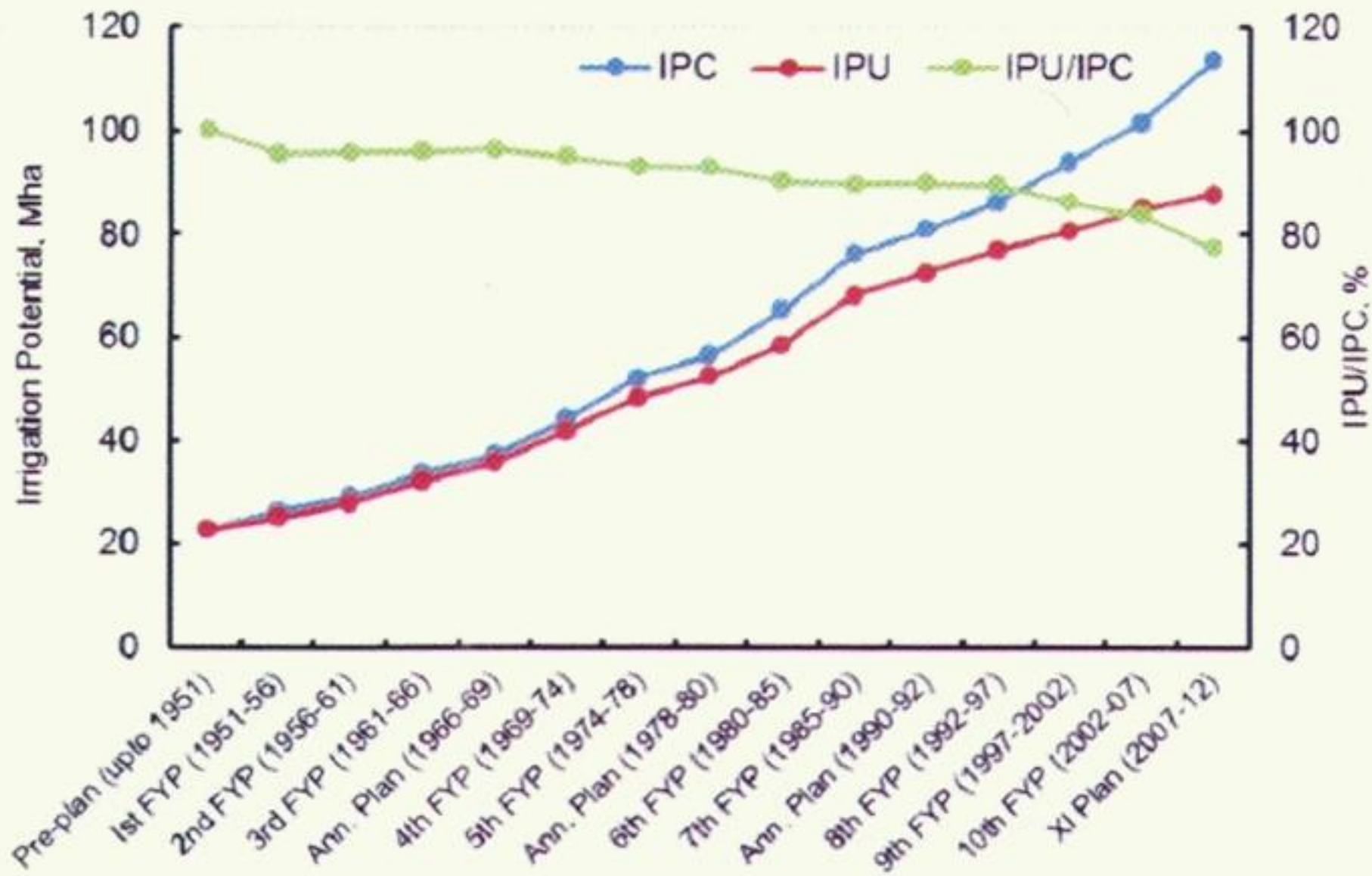
1 There is a huge gap between the **Irrigation Potential Created** and the **Irrigation Potential Utilized**.

2 Apart from the under-utilization, the efficiency of the irrigation systems is low at **30-38% for surface water**, **55% for ground water**.

3 Despite clear evidence of rising water stress, **water is still used inefficiently and indiscriminately**, particularly in agriculture.



STRATEGY FOR NEW INDIA @ 75



CONSTRAINTS

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WATER RESOURCES

9 The sustainability of the source and growing contamination of **ground water** are the constraints in ensuring safe drinking water supply in rural and urban areas.

7 Subsidized pricing of water has resulted in **non-revenue water** and a sharp decline in groundwater levels in all the States.

CONSTRAINTS IN UTILISATION OF WATER RESOURCES

8 As per 2011 Census, only **30.8% of the total rural households** and **70.6% of the total urban households** get piped water supply.



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WATER RESOURCES

WAY FORWARD

1. ON-GOING PROGRAMMES



- The water storage capacity is to be increased from the current level of 253 BCM to 304 BCM by completing the on-going programmes in time.
- Delays in the completion of PMKSY projects are to be looked at, and corrective measures should be taken.
- Command Area Development projects are essential to reduce the gap between irrigation potential created and utilized.
- The identified programmes, which are very important, are Ken-Betwa River linking project, the Pancheshwar project, the Rajasthan feeder and Sirhind feeders (Punjab & Rajasthan).
- National Mission for Clean Ganga needs to coordinate with different stakeholders.



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WATER RESOURCES

WAY FORWARD

2. WATER EFFICIENCY



- Incentivize wider adoption of sprinkler and drip irrigation by diverting resources, otherwise used to subsidize power and surface irrigation.
- Revitalize, renovate and repair water bodies.
- Special emphasis should be laid on desilting of water bodies, including rivers, lakes, ponds and reservoirs.

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WATER RESOURCES

WAY FORWARD

3. RECYCLING OF WASTE WATER



- With the country generating 140 BCM of wastewater annually, a pilot scheme to irrigate 10 lakh ha with treated waste water by 2020 may initially be taken up.
- Industries should be encouraged to meet a major share of their demand through recycled water.
- Moreover, programmes for smart water meters and tradable permits for use of recycled water may be launched.

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WATER RESOURCES

WAY FORWARD

4. GROUNDWATER MANAGEMENT



- There is a need to develop recharging zones at identified places to make groundwater resources sustainable by using check dams, farm ponds, tanks and injection wells.
- Participatory Aquifer Management, initiated in the 12th Plan and National Aquifer Management under PMKSY should be strengthened through a network of partnerships to control unbridled, competitive extraction of groundwater.

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WATER RESOURCES

WAY FORWARD

4. GROUNDWATER MANAGEMENT



Atal Bhujal Yojana

- Participatory approach to encourage **behavioral changes and community engagement** in ground water management at the Gram Panchayat level, as envisaged in the Atal Bhujal Yojana should be adopted and extended to other regions.
- **Promote the use of solar pumps** to improve the utilization of groundwater in Eastern India, where utilization is hampered by the lack of power.
- In Western India, **solar pumps with a buyback guarantee for surplus solar power** can offer reliable daytime energy for irrigation and stable cash income, as well as act as an incentive to conserve power and water.

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WATER RESOURCES

WAY FORWARD

4. GROUNDWATER MANAGEMENT



- PMKSY - Har Khet Ko Pani, envisaging enhancement of food production more than two-fold in 96 prioritized most deprived irrigation districts in 12 States, by creating irrigation facilities through tube wells, dug wells, bore wells and dug-cum-bore wells, should be expedited.
- This will facilitate assured irrigation in tribal and backward areas, that traditionally have been deprived of canal irrigation.
- Special focus should be kept on the quality of rural drinking water supply in arsenic and fluoride affected areas.

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WATER RESOURCES

WAY FORWARD

5. WATER HARVESTING

A. WATERSHED (CHECK DAM DEVELOPMENT)

- The Ministry of Jal Shakti may develop specific strategies to tap water through watershed development (check dams) in rain-fed areas, expand micro irrigation coverage to 80 lakh hectares, and link ground water development to aquifer mapping.
- The timeline for watershed development projects needs to be shortened with special efforts by State Governments.
- Funds available under MGNREGA may be used for watershed development.
- Introduce public-private partnerships in the water sector, initially to develop micro irrigation-based Command Area Development works based on a hybrid-annuity model.

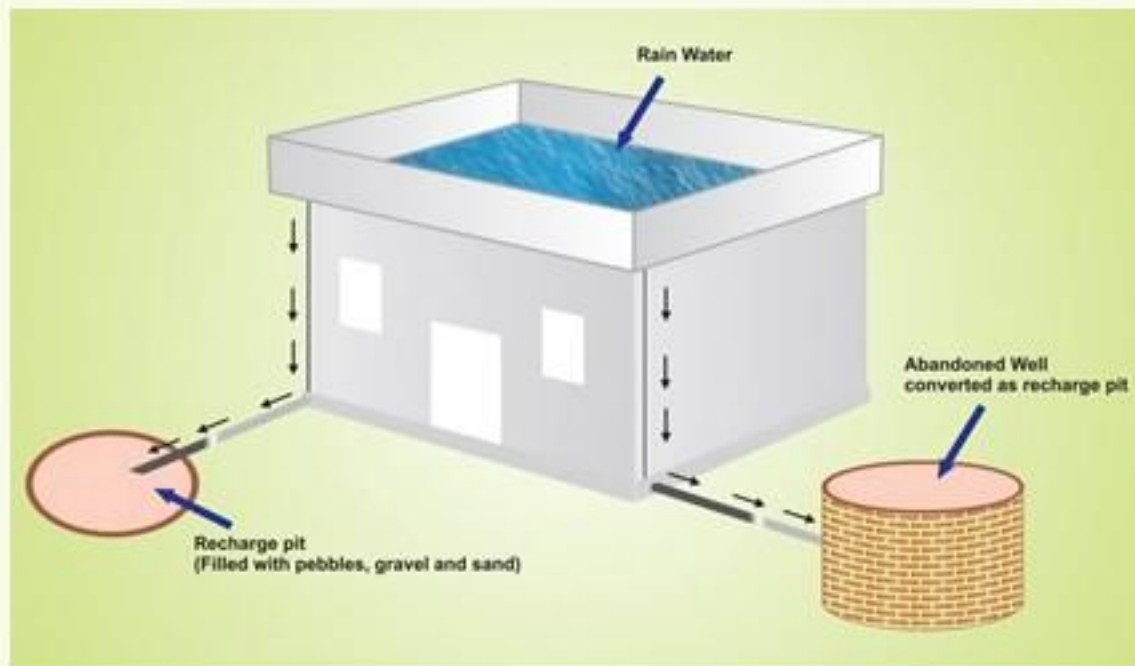


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WATER RESOURCES

WAY FORWARD

5. WATER HARVESTING



B. RAINWATER HARVESTING

- Model Building Bye Laws, 2016 circulated by Ministry of Urban Development, includes the provision of rainwater harvesting.
- Several states incorporated these in their respective Building Bye Laws.
- It is suggested that the states ensure effective implementation of the rainwater harvesting structures for buildings.

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WATER RESOURCES

WAY FORWARD

6. SUGGESTED REFORMS

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- Integrated River Basin Management approach is the need of the hour. River basin organizations for major basins may be expedited.
- NITI Aayog has developed a concept note on Revitalization of Rivers, which may be implemented on a pilot basis.
- To ensure Aviral and Nirmal Dhara in the Ganga, the river should be managed as a single system.
- Water regulatory framework should be established for water resources in all the States.
- An action plan should be drawn up to improve water use efficiency by 20% in all sectors by 2022.
- The Composite Water Management Index developed by NITI Aayog may be used as a tool to assess and improve the efficiency of water resources management.

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